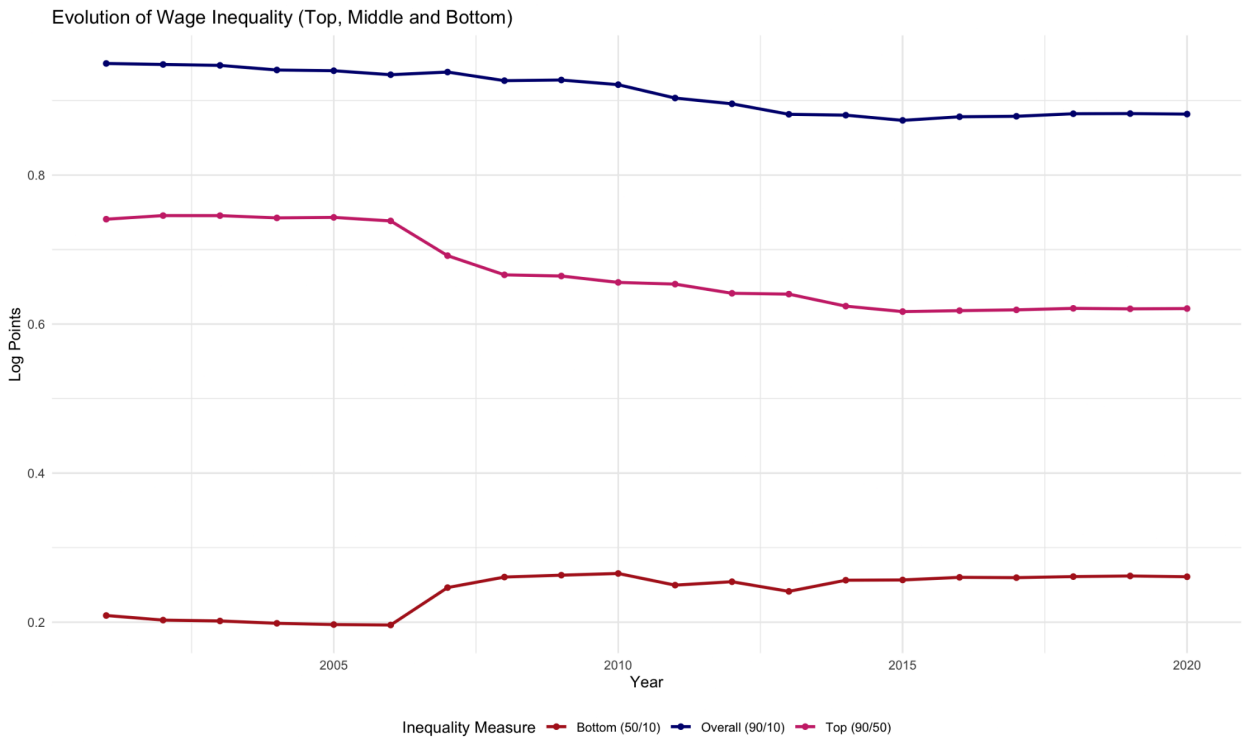
Unit 1

1 Wage Inequality (30%)

1.1 Aggregate Inequality (10%)

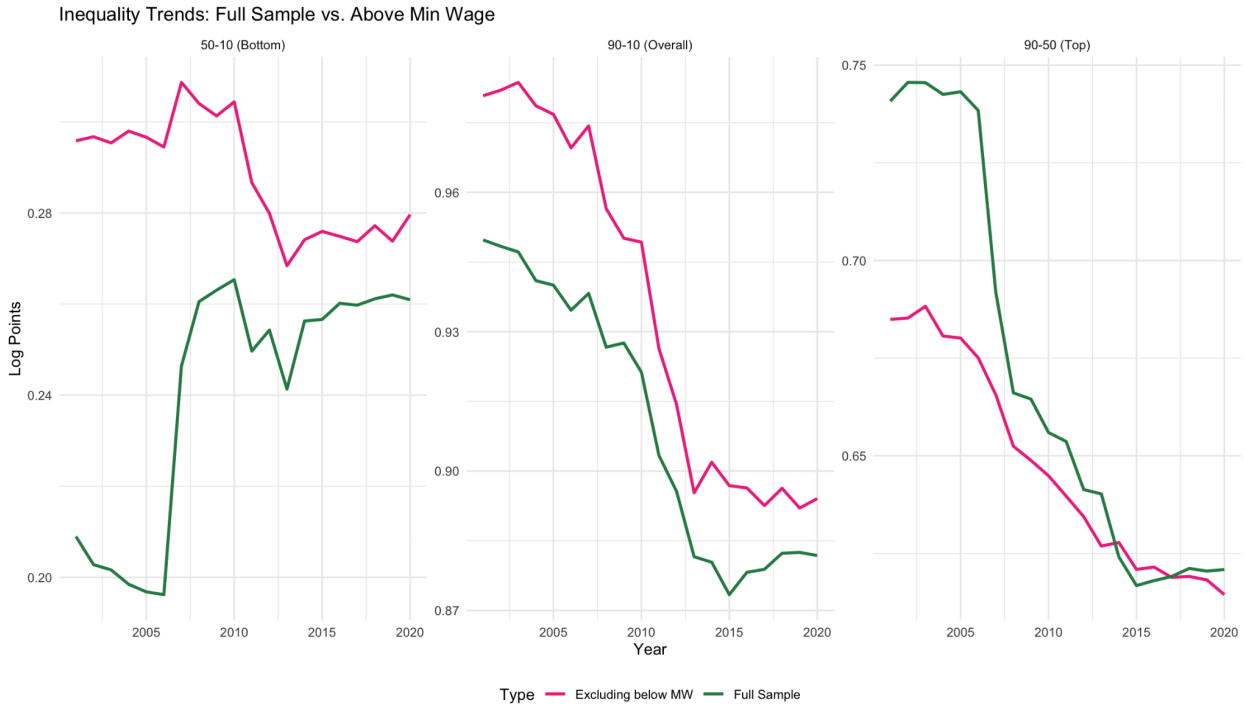
1. Compute the evolution of:
• log(p90)− log(p10)
• log(p50)− log(p10)
• log(p90)− log(p50)



Describe how inequality evolves over time. Are changes driven by movements at the top, bottom, or middle of the distribution?

The Overall (90/10) log wage gap falls from 0.950 in 2001 to 0.882 in 2020. This means aggregate wage inequality declines over time. The (50/10) gap rises from 0.209 to 0.261, while the (90/50) gap falls from 0.741 to 0.621 log points (now referred to as lp). The figure shows that inequality does not evolve uniformly across the wage distribution. The changes in aggregate inequality are predominantly driven by movements at the top of the distribution. Because the overall inequality measure (90/10) is the sum of the top (90/50) and bottom (50/10) measures, the direction of the aggregate trend depends on which component is more dominant. While the gap at the bottom widened by approximately 0.052 lp, this was more than offset by the larger 0.120 lp compression at the top. So, the narrowing of the (90/50) gap is the primary factor responsible for the overall reduction in wage dispersion observed in this period.

Does the pattern differ when excluding workers paid below the minimum wage? What does the difference tell you?

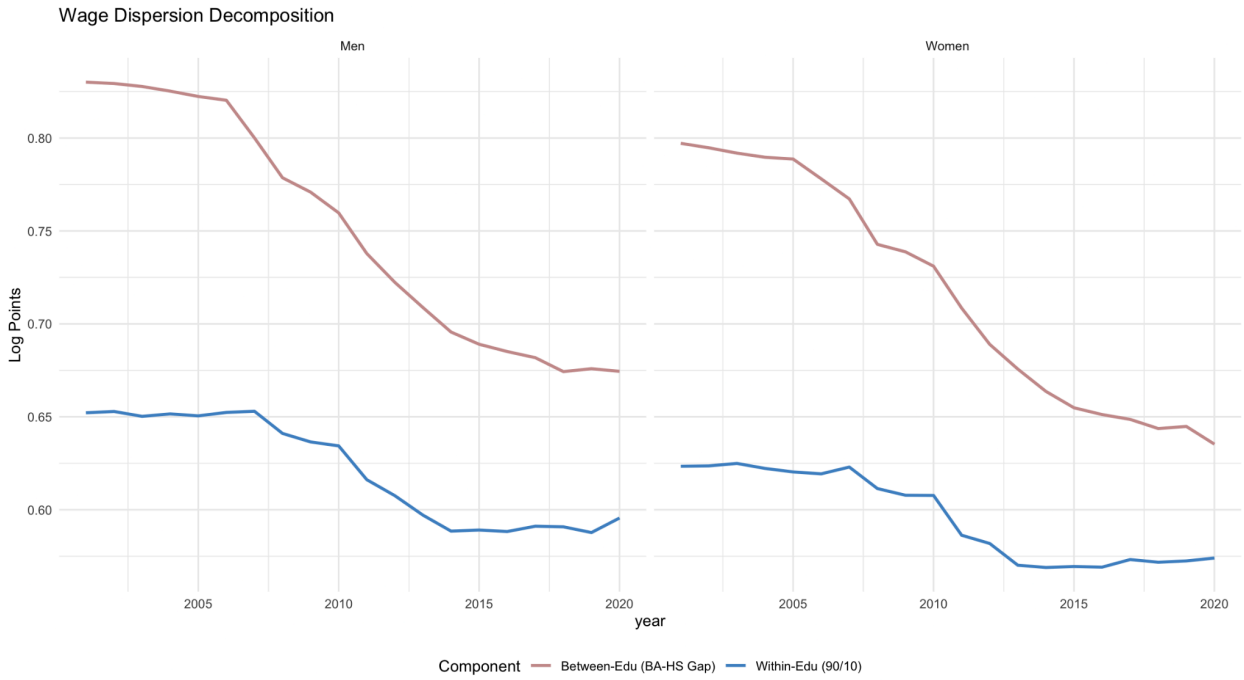


Yes, the pattern of wage inequality differs significantly when comparing the full sample and then Excluding Below Minimum Wages (ExBMW) group, particularly at the bottom of the distribution.

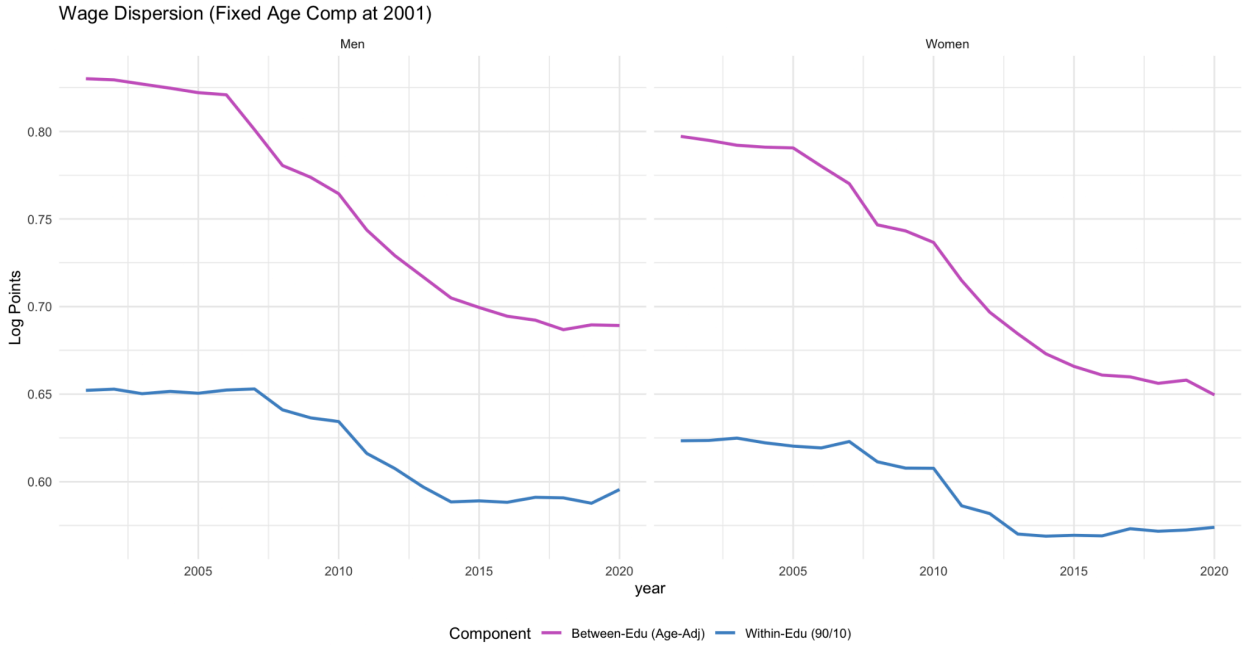
- *At the bottom end inequality (50/10):* In the full sample, inequality at the bottom increased over time and this is the only case where this has happened. The gap rose from approximately 0.21 to 0.26 lp, and this was a very sharp increase. In contrast when we consider the ExBMW group, they started initially much higher (around 0.30 lp) but followed a downward trend after around 2007, eventually narrowing to approximately 0.28 lp.
- *The overall inequality (90/10):* While both samples show a general decline in overall inequality, the ExBMW consistently exhibits a higher level of overall inequality than the whole sample. This suggests that the presence of sub-minimum wage workers in the full sample actually makes the 90/10 gap appear smaller than it is in the "compliant" labor market.
- *The top inequality (90/50):* Shows the most different trend between the two groups, with both experiencing a sharp and eventually overlapping decline. This confirms that the movements at the top of the distribution are largely independent of minimum wage enforcement issues.

1.2 Between and Within Group Inequality

1.2.1

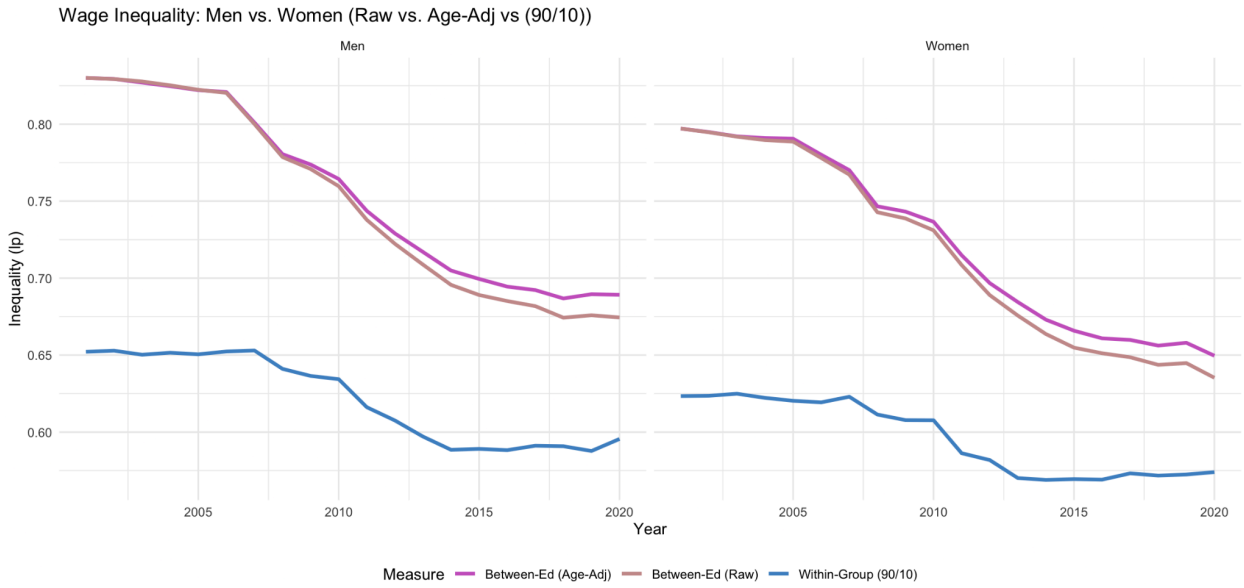


2. Generate the same graph holding the age composition of each education group constant at its base-year distribution.



Explain what this adjustment does and what it identifies.

It is easier to see the difference between Adjusted and Raw when the graphs are overlaid so we get:



The adjustment works by holding the age distribution of each education group constant at its 2001 (base-year) levels. Specifically, it calculates the mean log wage for each year by taking the observed wages for different age bins and weighting them according to the proportions observed in 2001. By stripping away the influence of an aging or young workforce, it reveals how the labor market's valuation of a BA degree versus a High School diploma has changed, independent of the experience levels of the people holding those degrees.

In particular, explain whether any divergence between the adjusted and unadjusted series reflects changes in the age composition within education groups over time, and what that reveals about cohort replacement.

There is a visible divergence where the adjusted series (Age-Adj) remains higher than the unadjusted (Raw) series for both men and women starting around 2007–2010. As the raw gap is lower than the age-adjusted gap, it indicates that the actual age composition within education groups over time was compressing the observed wage gap. If the age distribution had remained exactly as it was in 2001, the BA-HS wage gap would be larger today than it actually appears in the raw data.

As older, highly experienced (and so the higher-paid) BA holders retire, they are being replaced by a larger volume of younger BA holders who, despite having the degree, are at the lower-earning start of their careers. This trend exhibited by the BA workforce makes the education premium look smaller in the aggregate than it actually is when comparing workers of the same age. Therefore, the decline in the raw BA-HS gap is partially a demographic artifact of younger cohorts entering the market with degrees, rather than solely a decrease in the market value of education.

Which components account for changes in overall inequality?

While both Between and Within Education groups contributed to the overall compression of the wage distribution, the narrowing of the education premium (Between Group) was significantly more substantial and accounted for the bulk of the reduction in aggregate (90/10) inequality. This means most of the reduction in overall wage inequality comes from a narrowing of wage

differences across education groups rather than from lower residual dispersion within groups. Since the age-adjusted results are very similar, that conclusion does not seem to be driven by age composition.

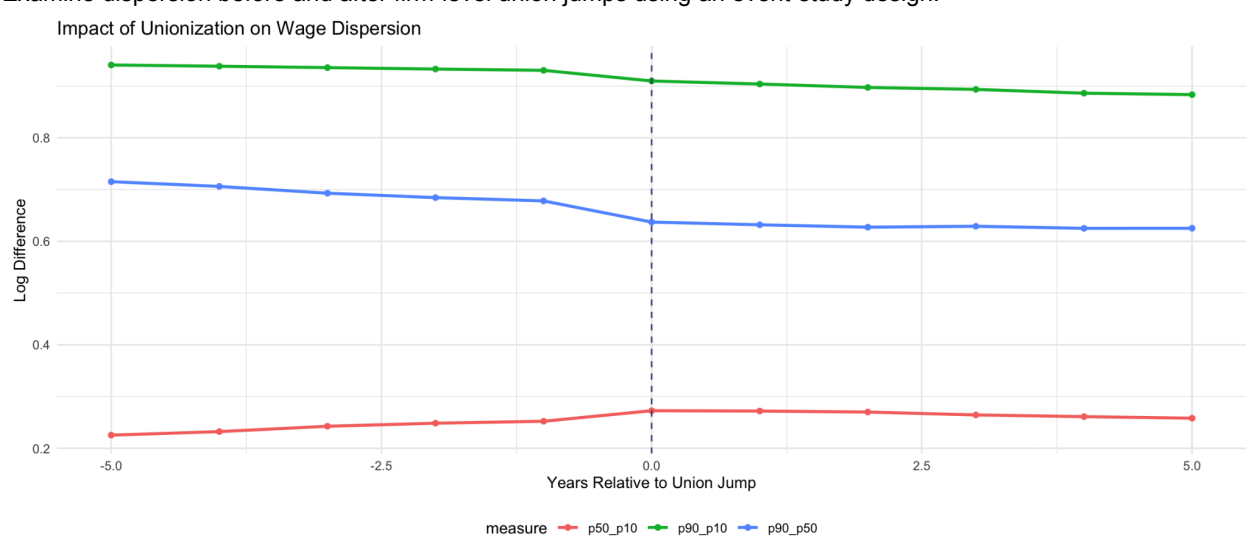
1.3 Role of Unions in Inequality

1. Compare wage dispersion across firms with different levels of union presence (use terciles of union coverage).

Union group	(overall) p90_p10	(upper tail) p90_p50	(lower tail) p50_p10
Low Union	1.020	0.706	0.315
Mid Union	0.969	0.685	0.283
High Union	0.763	0.530	0.233

Overall dispersion (90/10) falls monotonically from 1.02 to 0.76 as union presence increases, suggesting unions compress the wage distribution.
Both upper- and lower-tail dispersion follow the same pattern, but the lower-tail effect is smaller in magnitude (0.315 to 0.233) than the upper-tail (0.706 to 0.530).
This means unions compress wages more at the top of the distribution than the bottom, though both tails tighten. While the absolute compression is more pronounced at the top of the distribution (upper-tail), unions simultaneously narrow the gap at the bottom (lower-tail), leading to a much more egalitarian wage structure in high-union firms.

Examine dispersion before and after firm-level union jumps using an event-study design.



Here, year 0 is the event year and not a calendar year. It represents the year the union was formed and -5 is five years before the union was formed and 5 is five years after the union was formed.

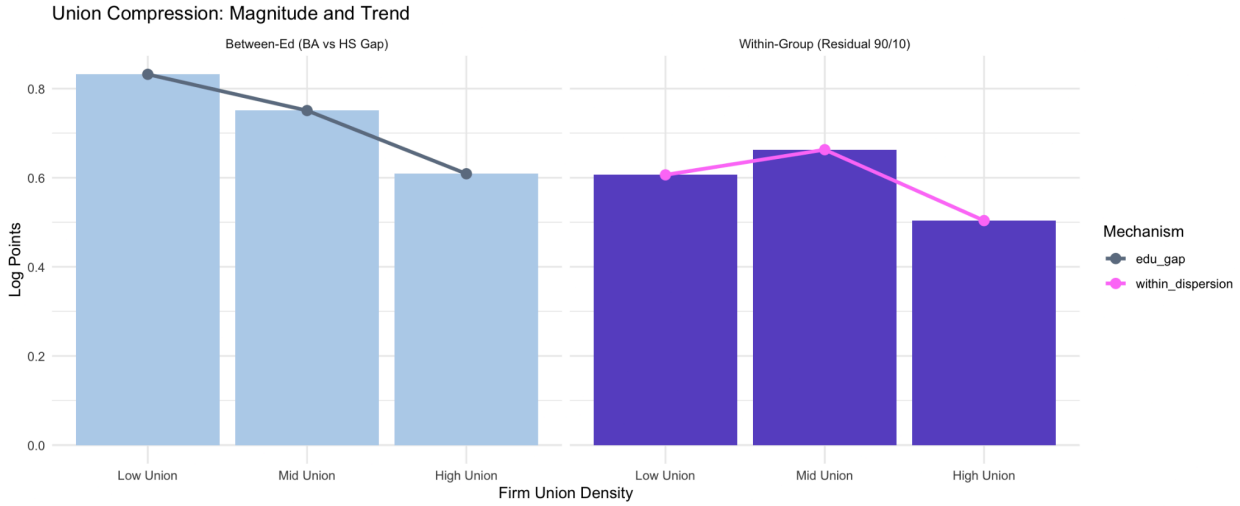
- Overall Inequality (90/10): Aggregate inequality was relatively stable or slightly declining prior to the jump but experienced a sustained decrease starting at Year 0, falling from approximately 0.93 to 0.88 lp by Year 5.
- Upper-Tail Inequality (90/50): There is a sharp drop in the (90/50) gap at the moment of the union jump, which continues to compress from roughly 0.68 to 0.62 lp over the following five years.
- Lower-Tail Inequality (50/10): Inequality at the bottom was actually increasing slightly prior to the jump (from ~0.23 to ~0.26). After the unionization event, this upward trend reverses and flattens, ending at a slightly lower level of dispersion.

Discuss whether pre-trends are present and what they imply for identification.

There are visible yet limited pre-trends, particularly in the 90/50 (upper-tail) and 90/10 (overall) measures. These gaps were already narrowing slightly between Year -5 and Year -1, suggesting that firms experiencing union jumps were not on a perfectly stable trajectory prior to the unionization event. Despite the pre-trends, the change in the slope (the acceleration of compression) at Year 0 for the (90/50) gap and the reversal of the trend for the (50/10) gap provide evidence that unionization exerts an independent compressing force on the wage distribution beyond existing trends.

Identification: The existence of these pre-trends suggests a possible violation of the parallel trends assumption. If firms that unionize were already moving toward more compressed wage structures for other reasons (like falling productivity or changing workforce composition) then the event-study estimates may overstate the causal impact of the union itself.

Does unionization compress within-firm wage dispersion, between-group wage gaps, or both? Provide an economic interpretation of your findings.



Here, unionization compresses both within-firm wage dispersion and between-group wage gaps. First, they narrow the gap between educated and less-educated workers. In low-union firms, a worker with a BA degree earns about 0.83 lp more than a HS graduate. In high-union

firms, that gap shrinks to 0.61. Unions essentially reduce the "reward" for having a degree, they push for equal treatment regardless of credentials. Within-group dispersion actually rises slightly in mid-union firms before falling sharply in high-union firms. This suggests the compressing effect on within-firm dispersion only kicks in at sufficiently high union density, whereas the education gap compresses more gradually and consistently. Intuitively, unions push for standardised, rule-based pay which leaves less room for individual characteristics, whether education, skill, etc, to drive wage differences.

Appendix for U1

Section 1, agg ineq

Year	p10	p50	p90	Overall Inequality	Bottom Inequality	Top Inequality
2001	7.61	9.38	19.7	0.950	0.209	0.741
2002	7.65	9.37	19.8	0.948	0.203	0.746
2003	7.67	9.38	19.8	0.947	0.202	0.746
2004	7.70	9.40	19.7	0.941	0.198	0.743
2005	7.73	9.41	19.8	0.940	0.197	0.743
2006	7.77	9.46	19.8	0.935	0.196	0.738
2007	7.78	9.95	19.9	0.938	0.246	0.692
2008	7.85	10.2	19.8	0.927	0.261	0.666
2009	7.89	10.3	19.9	0.928	0.263	0.665
2010	7.93	10.3	19.9	0.921	0.265	0.656
2011	8.10	10.4	20.0	0.903	0.250	0.654
2012	8.17	10.5	20.0	0.896	0.254	0.641
2013	8.32	10.6	20.1	0.882	0.241	0.640
2014	8.35	10.8	20.1	0.880	0.256	0.624
2015	8.40	10.9	20.1	0.873	0.257	0.617
2016	8.41	10.9	20.2	0.878	0.260	0.618
2017	8.43	10.9	20.3	0.879	0.260	0.619
2018	8.44	11.0	20.4	0.882	0.261	0.621
2019	8.46	11.0	20.4	0.882	0.262	0.620
2020	8.49	11.0	20.5	0.882	0.261	0.621

1.1.3

#	year	p10	p50	p90	overall	bottom	top	Type
1	2001	7.61	9.38	19.7	0.950	0.209	0.741	Full Sample
2	2002	7.65	9.37	19.8	0.948	0.203	0.746	Full Sample
3	2003	7.67	9.38	19.8	0.947	0.202	0.746	Full Sample
4	2004	7.70	9.40	19.7	0.941	0.198	0.743	Full Sample
5	2005	7.73	9.41	19.8	0.940	0.197	0.743	Full Sample
6	2006	7.77	9.46	19.8	0.935	0.196	0.738	Full Sample
7	2007	7.78	9.95	19.9	0.938	0.246	0.692	Full Sample
8	2008	7.85	10.2	19.8	0.927	0.261	0.666	Full Sample
9	2009	7.89	10.3	19.9	0.928	0.263	0.665	Full Sample
10	2010	7.93	10.3	19.9	0.921	0.265	0.656	Full Sample
11	2011	8.10	10.4	20.0	0.903	0.250	0.654	Full Sample
12	2012	8.17	10.5	20.0	0.896	0.254	0.641	Full Sample
13	2013	8.32	10.6	20.1	0.882	0.241	0.640	Full Sample
14	2014	8.35	10.8	20.1	0.880	0.256	0.624	Full Sample
15	2015	8.40	10.9	20.1	0.873	0.257	0.617	Full Sample
16	2016	8.41	10.9	20.2	0.878	0.260	0.618	Full Sample
17	2017	8.43	10.9	20.3	0.879	0.260	0.619	Full Sample
18	2018	8.44	11.0	20.4	0.882	0.261	0.621	Full Sample
19	2019	8.46	11.0	20.4	0.882	0.262	0.620	Full Sample
20	2020	8.49	11.0	20.5	0.882	0.261	0.621	Full Sample
21	2001	7.95	10.7	21.2	0.981	0.296	0.685	Excluding BMW
22	2002	7.97	10.7	21.3	0.982	0.297	0.685	Excluding BMW
23	2003	7.98	10.7	21.3	0.984	0.295	0.688	Excluding BMW

24	2004	7.99	10.8	21.3	0.979	0.298	0.681	Excluding BMW
25	2005	8.02	10.8	21.3	0.977	0.297	0.680	Excluding BMW
26	2006	8.07	10.8	21.3	0.970	0.294	0.675	Excluding BMW
27	2007	8.09	11.0	21.4	0.974	0.309	0.666	Excluding BMW
28	2008	8.28	11.2	21.5	0.956	0.304	0.652	Excluding BMW
29	2009	8.32	11.2	21.5	0.950	0.301	0.649	Excluding BMW
30	2010	8.34	11.3	21.5	0.949	0.304	0.645	Excluding BMW
31	2011	8.57	11.4	21.6	0.926	0.287	0.640	Excluding BMW
32	2012	8.70	11.5	21.7	0.914	0.280	0.634	Excluding BMW
33	2013	8.89	11.6	21.8	0.895	0.268	0.627	Excluding BMW
34	2014	8.88	11.7	21.9	0.902	0.274	0.628	Excluding BMW
35	2015	8.92	11.8	21.9	0.897	0.276	0.621	Excluding BMW
36	2016	8.96	11.8	22.0	0.896	0.275	0.621	Excluding BMW
37	2017	9.00	11.8	22.0	0.893	0.274	0.619	Excluding BMW
38	2018	8.99	11.9	22.0	0.896	0.277	0.619	Excluding BMW
39	2019	9.07	11.9	22.1	0.892	0.274	0.618	Excluding BMW
40	2020	9.08	12.0	22.2	0.894	0.280	0.614	Excluding BMW

Unit 2

2 Do Firms Matter for Inequality?

2.1 Firm Wage Premia

This exercise involves AKM decompositions.

What fraction of total wage variance is explained by worker fixed effects, firm fixed effects, and their covariance?

Period	share_worker	share_firm	share_cov
2001–2010	0.185 (18.5%)	0.172 (17.2%)	0.00813
2011–2020	0.215 (21.5%)	0.167 (16.7%)	0.00325

The importance of worker heterogeneity increased by 3 percentage points, while the contribution of firm premiums slightly declined by 0.5 percentage points. Notably, the contribution of sorting, that is the tendency of high-ability workers to work at high-paying firms, is extremely small (0.00813) and actually decreased (to 0.00325) in the second period. Which is interesting as it is usually expected to be higher.

Discuss what additional assumptions are needed to make sub-period estimates comparable.

To estimate the model separately on sub-periods and compare them, several additional assumptions are required:

- One must assume that worker and firm effects are constant over the duration of each specific sub-period.
- Identification relies on the assumption that worker moves between firms are "exogenous," meaning they are not driven by the match-specific or transitory error components of the wage.
- The estimates are only comparable if they are performed on the largest connected set of workers and firms for each period, that is the subset reachable through chains of worker moves. If the composition of these sets changes significantly, the relative shares may reflect changes in labor market connectivity rather than true wage structure shifts.

Compare your variance decomposition with the results reported in Card, Heining, and Kline (2013, QJE) for West Germany. Are there important differences?

The comparison with Card, Heining, and Kline (CHK, 2013) suggests some similarities and differences. It is important to note that firm effects matter in both cases. CHK found that the variability of establishment-level pay premiums increased significantly, accounting for about 25% of the total rise in wage variance (unlike the results here which also account for a significant portion but decrease to 16.7% in the next period).

CHK revealed that rise in assortative matching (covariance) was a dominant driver of inequality, with its share of total variance jumping from 2.3% to 16.4%. In this study, sorting is almost insignificant at 0.00325 in the second period after falling from 0.00813.

Is there evidence of assortative matching between workers and firms? Compute the correlation between estimated worker and firm fixed effects among movers.

Correlation (Movers only) 2001-2010	0.0051
Correlation (Movers only) 2011-2020	- 0.004

These near-zero values indicate that "high-ability" workers (as defined by their unobserved wage component) are not systematically sorted into "high-paying" firms. This stands in sharp contrast to findings in CHK's paper as this correlation was found to have risen significantly to approximately 0.25.

Discuss the potential for bias in this estimate (the limited mobility bias) and whether it is likely to affect your conclusions.

- The limited mobility bias is a known technical challenge in AKM estimation where sampling errors in the fixed effects tend to be negatively correlated.
- As firm effects are identified only through workers who move between firms, a small number of movers leads to noisy estimates. This noise generally biases the correlation downward (by making it appear more negative than it truly is).
- While this bias suggests the "true" correlation might be slightly more positive than the -0.004 recorded in the second period, it does not change the fundamental conclusion that sorting is not a major driver of inequality in this dataset.
- Even if the bias were corrected, the correlation would likely remain far below the levels observed in the West German study.

Compute the correlation between observable worker characteristics and firm effects. Does it surprise you?

Corr (Observables, Firm Effects) 2001-2010	0.1206
Corr (Observables, Firm Effects) 2011-2020	0.1294

This result is not particularly surprising because it suggests a mismatch in how firms sort workers. While firms do not appear to sort based on the "portable" unobserved ability of workers they do show a modest tendency to sort based on demographics like age and experience.
This implies that older or more experienced workers are slightly more likely to be found in firms that pay higher baseline premiums, whereas unobservables remain randomly distributed across firms.
So the overall picture is quite clear: there is little evidence of sorting of workers, but there is some positive sorting on observables.

Are firm wage premia related to unionization? Regress estimated firm fixed effects on firm-level union presence and interpret the result. Does the direction of this relationship surprise you?

Union Effect on Firm Premium (2001-2010)	-0.8763
Union Effect on Firm Premium (2011-2020)	-0.6624

Yes, firm wage premia is significantly and inversely related to unionization. A negative coefficient indicates that firms with higher union density tend to have lower estimated wage premiums. This means that, holding worker characteristics and individual ability constant, the baseline pay level at a highly unionized firm is lower than at a firm with low union presence. While the magnitude of this negative relationship decreased slightly in the second period, it remains a strong and statistically relevant association.

Decompose total wage dispersion into between-firm and within-firm components.

Period	total_variance	between_share	within_share
2001–2010	0.148	0.253 (25.3%)	0.750 (75%)
2011–2020	0.124	0.260 (26%)	0.743 (74.3%)

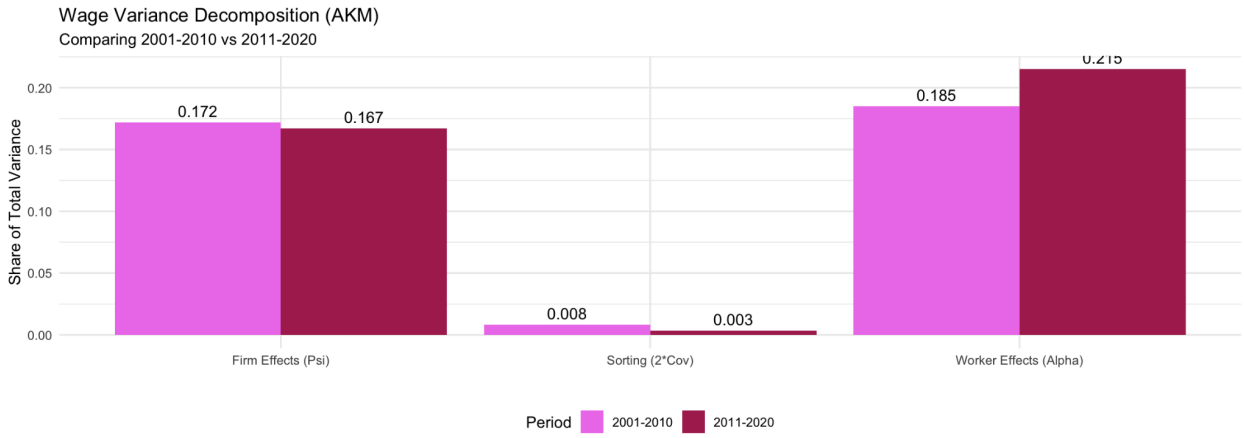
Which component accounts for observed changes in inequality over time, and does the union reform alter their relative importance?

Total wage variance fell from 0.148 to 0.124 across the two decades. Within-firm dispersion dominates throughout, explaining roughly three-quarters of total inequality in both periods. This means that most wage differences exist between workers at the same firm, not across firms.

The slight shift in shares (within falling from 75% to 74.3%, between rising from 25.3% to 26%) suggests the decline in overall inequality was driven primarily by compression inside firms rather than firms converging toward each other.

Unionization likely contributed to this. High-union firms show meaningfully lower within-firm dispersion and a narrower education premium than low-union firms. This is consistent with unions standardising pay scales internally (reducing the spread between high and low earners within the same establishment) while having slightly lesser effect on differences across firms. Both channels compress inequality, but the within-firm channel dominates.

Appendix for U2


Wage Variance Decomposition (AKM)
Comparing 2001-2010 vs 2011-2020
Share of Total Variance
0.20
0.15
0.10
0.05
0.00
0.172
0.167
0.008
0.003
0.185
0.215
Firm Effects (Psi)
Sorting (2*Cov)
Worker Effects (Alpha)
Period
2001-2010
2011-2020

Unit 3

3 Minimum Wages, Unions, and Non-Compliance

3.1 Estimation

It is important to note that in this dataset, "Female" (the gender dummy) shows that workers have changed gender over time. For this reason, "Female" has been included in all regression equations as a control for within-worker variation. Similarly, "Age" is included as a control in all specifications, including those with state×year fixed effects, as it varies at the individual level and is therefore not collinear with state×year dummies. This allows lifecycle wage effects to be absorbed across all specifications.

(a) No fixed effects (OLS with worker controls only)

Model 1 — Log Wages (employed workers only):

$$\log(w_i) = \beta_0 + \beta_1 \text{MinWage}_i + \beta_2 \text{Age}_i + \beta_3 \text{Female}_i + \beta_4 \text{Education}_i + \varepsilon_i$$

Model 2 — Employment Probability (full sample):

$$\text{Employed}_i = \beta_0 + \beta_1 \text{MinWage}_i + \beta_2 \text{Age}_i + \beta_3 \text{Female}_i + \beta_4 \text{Education}_i + \varepsilon_i$$

Model 3 — Non-Compliance (employed workers only):

$$\text{NonCompliant}_i = \beta_0 + \beta_1 \text{MinWage}_i + \beta_2 \text{Age}_i + \beta_3 \text{Female}_i + \beta_4 \text{Education}_i + \varepsilon_i$$

(b) Firm fixed effects

Model 1 — Log Wages (employed workers only):

$$\log(w_{it}) = \beta_1 \text{MinWage}_{it} + \beta_2 \text{Age}_{it} + \beta_3 \text{Female}_i + \beta_4 \text{Education}_i + \alpha_j + \varepsilon_{it}$$

Model 2 — Employment Probability (full sample):

$$\text{Employed}_{it} = \beta_1 \text{MinWage}_{it} + \beta_2 \text{Age}_{it} + \beta_3 \text{Female}_i + \beta_4 \text{Education}_i + \alpha_j + \varepsilon_{it}$$

Model 3 — Non-Compliance (employed workers only):

$$\text{NonCompliant}_{it} = \beta_1 \text{MinWage}_{it} + \beta_2 \text{Age}_{it} + \beta_3 \text{Female}_i + \beta_4 \text{Education}_i + \alpha_j + \varepsilon_{it}$$

where α_j denotes firm fixed effects.

(c) Worker and firm fixed effects

Model 1 — Log Wages (employed workers only):

$$\log(w_{it}) = \beta_1 \text{MinWage}_{it} + \beta_2 \text{Age}_{it} + \beta_3 \text{Female}_i + \beta_4 \text{Education}_i + \alpha_i + \alpha_j + \varepsilon_{it}$$

Model 2 — Employment Probability (full sample):

$$\text{Employed}_{it} = \beta_1 \text{MinWage}_{it} + \beta_2 \text{Age}_{it} + \beta_3 \text{Female}_i + \beta_4 \text{Education}_i + \alpha_i + \alpha_j + \varepsilon_{it}$$

Model 3 — Non-Compliance (employed workers only):

$$\text{NonCompliant}_{it} = \beta_1 \text{MinWage}_{it} + \beta_2 \text{Age}_{it} + \beta_3 \text{Female}_i + \beta_4 \text{Education}_i + \alpha_i + \alpha_j + \varepsilon_{it}$$

where α_i denotes worker fixed effects and α_j denotes firm fixed effects.

(d) Worker, firm, and state×year fixed effects

Model 1 — Log Wages (employed workers only):

$$\log(w_{it}) = \beta_1 \text{Age}_{it} + \beta_2 \text{Female}_i + \beta_3 \text{Education}_i + \alpha_i + \alpha_j + \alpha_{s,t} + \varepsilon_{it}$$

Model 2 — Employment Probability (full sample):

$$\text{Employed}_{it} = \beta_1 \text{Age}_{it} + \beta_2 \text{Female}_i + \beta_3 \text{Education}_i + \alpha_i + \alpha_j + \alpha_{s,t} + \varepsilon_{it}$$

Model 3 — Non-Compliance (employed workers only):

$$\text{NonCompliant}_{it} = \beta_1 \text{Age}_{it} + \beta_2 \text{Female}_i + \beta_3 \text{Education}_i + \alpha_i + \alpha_j + \alpha_{s,t} + \varepsilon_{it}$$

where α_i = worker FE, α_j = firm FE, $\alpha_{s,t}$ = state×year FE. Note: `min_wage` is excluded due to collinearity with $\alpha_{s,t}$.

Specification	$\hat{\beta}_1$ (Log Wage)	$\hat{\beta}_1$ (Employment)	$\hat{\beta}_1$ (Non-Compliance)
a) OLS	0.0452*** (0.0002)	−0.0028*** (0.0002)	0.0449*** (0.0003)
b) Firm FE	0.0485*** (0.0006)	−0.0020*** (0.0003)	0.0405*** (0.0005)

c) Worker + Firm FE	0.0489*** (0.0003)	−0.0020*** (0.0003)	0.0409*** (0.0004)
d) Worker + Firm + State×Year FE	*collinear — dropped*	*collinear — dropped*	*collinear — dropped*

The minimum wage coefficient is stable across all three specifications. For log wages, $\hat{\beta}_1$ rises marginally from 0.0452 (OLS) to 0.0489 (worker + firm FE), roughly a 4.9% wage increase per dollar rise in the minimum wage. The stability across specifications suggests OLS was not severely biased by firm-level sorting or worker heterogeneity, lending credibility to the causal interpretation. It is also important to note here that the FE absorb the constant term fully.

The employment effect is negative but economically negligible (−0.0028 to −0.0020). Despite being statistically significant (possibly because of the large sample size) the magnitude suggests minimum wage increases do not meaningfully reduce employment.

The non-compliance result is important to note. It shows that higher minimum wages increase the share of workers paid below the legal floor. This likely reflects a compliance cost channel, as the minimum rises, low-margin firms struggle to adjust, widening the gap between the legal requirement and actual wages.

In specification (d), min_wage is dropped entirely due to perfect collinearity with state×year fixed effects, since minimum wages vary only at the state-year level. The code was written also to check how minimum wage changed as per every state-year combination and the result was that it stayed constant. Which means all the variation between minimum wages can be fully explained by changes in the state-year level.

Step 2: Adding unions and the interaction.

(a) No fixed effects

Model 1 — Log Wages (employed workers only):

$$\log(w_i) = \beta_0 + \beta_1 \text{MinWage}_i + \beta_2 \text{Union}_i + \beta_3 (\text{MinWage} \times \text{Union})_i + \beta_4 \text{Age}_i + \beta_5 \text{Female}_i + \beta_6 \text{Education}_i + \varepsilon_i$$

Model 2 — Employment Probability (full sample):

$$\text{Employed}_i = \beta_0 + \beta_1 \text{MinWage}_i + \beta_2 \text{Union}_i + \beta_3 (\text{MinWage} \times \text{Union})_i + \beta_4 \text{Age}_i + \beta_5 \text{Female}_i + \beta_6 \text{Education}_i + \varepsilon_i$$

Model 3 — Non-Compliance (employed workers only):

$$\text{NonCompliant}_i = \beta_0 + \beta_1 \text{MinWage}_i + \beta_2 \text{Union}_i + \beta_3 (\text{MinWage} \times \text{Union})_i + \beta_4 \text{Age}_i + \beta_5 \text{Female}_i + \beta_6 \text{Education}_i + \varepsilon_i$$

(b) Firm fixed effects

Model 1 — Log Wages (employed workers only):

$$\log(w_{it}) = \beta_1 \text{MinWage}_{it} + \beta_2 \text{Union}_{it} + \beta_3 (\text{MinWage} \times \text{Union})_{it} + \beta_4 \text{Age}_{it} + \beta_5 \text{Female}_i + \beta_6 \text{Education}_i + \alpha_j + \varepsilon_{it}$$

Model 2 — Employment Probability (full sample):

$$\text{Employed}_{it} = \beta_1 \text{MinWage}_{it} + \beta_2 \text{Union}_{it} + \beta_3 (\text{MinWage} \times \text{Union})_{it} + \beta_4 \text{Age}_{it} + \beta_5 \text{Female}_i + \beta_6 \text{Education}_i + \alpha_j + \varepsilon_{it}$$

Model 3 — Non-Compliance (employed workers only):

$$\text{NonCompliant}_{it} = \beta_1 \text{MinWage}_{it} + \beta_2 \text{Union}_{it} + \beta_3 (\text{MinWage} \times \text{Union})_{it} + \beta_4 \text{Age}_{it} + \beta_5 \text{Female}_i + \beta_6 \text{Education}_i + \alpha_j + \varepsilon_{it}$$

where α_j denotes firm fixed effects.

(c) Worker and firm fixed effects

Model 1 — Log Wages (employed workers only):

$$\log(w_{it}) = \beta_1 \text{MinWage}_{it} + \beta_2 \text{Union}_{it} + \beta_3 (\text{MinWage} \times \text{Union})_{it} + \beta_4 \text{Age}_{it} + \beta_5 \text{Female}_i + \beta_6 \text{Education}_i + \alpha_i + \alpha_j + \varepsilon_{it}$$

Model 2 — Employment Probability (full sample):

$$\text{Employed}_{it} = \beta_1 \text{MinWage}_{it} + \beta_2 \text{Union}_{it} + \beta_3 (\text{MinWage} \times \text{Union})_{it} + \beta_4 \text{Age}_{it} + \beta_5 \text{Female}_i + \beta_6 \text{Education}_i + \alpha_i + \alpha_j + \varepsilon_{it}$$

Model 3 — Non-Compliance (employed workers only):

$$\text{NonCompliant}_{it} = \beta_1 \text{MinWage}_{it} + \beta_2 \text{Union}_{it} + \beta_3 (\text{MinWage} \times \text{Union})_{it} + \beta_4 \text{Age}_{it} + \beta_5 \text{Female}_i + \beta_6 \text{Education}_i + \alpha_i + \alpha_j + \varepsilon_i$$

where α_i denotes worker fixed effects and α_j denotes firm fixed effects.

(d) Worker, firm, and state×year fixed effects

Model 1 — Log Wages (employed workers only):

$$\log(w_{it}) = \beta_2 \text{Union}_{it} + \beta_3 (\text{MinWage} \times \text{Union})_{it} + \beta_4 \text{Age}_{it} + \beta_5 \text{Female}_i + \beta_6 \text{Education}_i + \alpha_i + \alpha_j + \alpha_{s,t} + \varepsilon_{it}$$

Model 2 — Employment Probability (full sample):

$$\text{Employed}_{it} = \beta_2 \text{Union}_{it} + \beta_3 (\text{MinWage} \times \text{Union})_{it} + \beta_4 \text{Age}_{it} + \beta_5 \text{Female}_i + \beta_6 \text{Education}_i + \alpha_i + \alpha_j + \alpha_{s,t} + \varepsilon_{it}$$

Model 3 — Non-Compliance (employed workers only):

$$\text{NonCompliant}_{it} = \beta_2 \text{Union}_{it} + \beta_3 (\text{MinWage} \times \text{Union})_{it} + \beta_4 \text{Age}_{it} + \beta_5 \text{Female}_i + \beta_6 \text{Education}_i + \alpha_i + \alpha_j + \alpha_{s,t} + \varepsilon_{it}$$

where α_i = worker FE, α_j = firm FE, $\alpha_{s,t}$ = state×year FE. Note: `min_wage` dropped due to collinearity with $\alpha_{s,t}$; the interaction term is retained as union status varies within state-year cells.

Specification	Log Wage	Employment	Non-Compliance
Panel A: Min Wage ($\hat{\beta}_1$)			
a) No FE	0.0413*** (0.0004)	−0.0113*** (0.0004)	0.0540*** (0.0006)
b) Firm FE	0.0414*** (0.0007)	−0.0149*** (0.0006)	0.0550*** (0.0008)
c) Worker + Firm FE	0.0419*** (0.0004)	−0.0150*** (0.0006)	0.0551*** (0.0008)
d) Worker + Firm + State×Year FE	*dropped (collinear)*	*dropped (collinear)*	*dropped (collinear)*
Panel B: Union ($\hat{\beta}_2$)			
a) No FE	0.0228*** (0.0037)	−0.0561*** (0.0045)	0.0146* (0.0059)
b) Firm FE	0.0212*** (0.0053)	−0.0683*** (0.0051)	0.0234** (0.0072)
c) Worker + Firm FE	0.0227*** (0.0036)	−0.0693*** (0.0051)	0.0236*** (0.0068)
d) Worker + Firm + State×Year FE	0.0190*** (0.0041)	−0.0744*** (0.0080)	0.0072 (0.0100)
Panel C: Min Wage × Union ($\hat{\beta}_3$)			
a) No FE	−0.0004 (0.0004)	0.0081*** (0.0005)	−0.0047*** (0.0007)
b) Firm FE	−0.0003 (0.0006)	0.0097*** (0.0006)	−0.0058*** (0.0008)
c) Worker + Firm FE	−0.0005 (0.0004)	0.0098*** (0.0006)	−0.0057*** (0.0008)
d) Worker + Firm + State×Year FE	−0.0001 (0.0005)	0.0105*** (0.0009)	−0.0039*** (0.0012)

Marginal Effect at Mean MW	0.01847194
Marginal Effect at Max MW	0.01729382

$\hat{\beta}_1$ is stable across specifications (a)–(c): for log wages it moves from 0.0413 to 0.0419, employment from −0.0113 to −0.0150, and non-compliance from 0.0540 to 0.0551 — suggesting these effects are not driven by firm or worker sorting. In specification (d), $\hat{\beta}_1$ is

dropped due to perfect collinearity with state×year FEs, but the interaction term survives since union status varies within state-year cells. This results are in line with the ones from Step 1.

$\hat{\beta}_2$ alone does not summarise the union effect once the interaction is included. The total marginal effect $\partial Y/\partial \text{Union} = \hat{\beta}_2 + \hat{\beta}_3 \times \text{MW}$ evaluates to 0.0185 at the mean and 0.0173 at the maximum minimum wage which means that union workers earn roughly 1.8–1.9% more, but the premium shrinks as MW rises. This is intuitive: at high minimum wage levels, the statutory floor already compresses wages mechanically, leaving less room for unions to add further compression.

For employment, $\hat{\beta}_3$ is positive and stable (~0.0098), meaning unions partially offset the disemployment effects of minimum wages through collective agreements stabilising worker-firm matches.

For non-compliance, $\hat{\beta}_3$ is negative (−0.0047 to −0.0057), and the total union effect at MW=10 is −0.0337, reflecting unions directly enforcing wage floors through collective bargaining rather than relying on regulatory oversight alone.

In specification (d) with state×year fixed effects, β_1 is no longer identified but β_3 remains identified. Explain precisely why. What does λ_{st} absorb, and what source of variation does β_3 exploit that β_1 cannot?

The state×year fixed effect absorbs all variation that is common to workers within the same state and year — including the minimum wage itself, since MW is set at the state-year level. Every worker in a given state-year faces the same MW, so once state×year is included, there is no remaining within-cell variation in MW to identify $\hat{\beta}_1$. It is perfectly collinear and dropped.

$\hat{\beta}_3$ survives because it exploits a different source of variation. The interaction term MW × Union varies *within* state-year cells — two workers in the same state and year can differ in union status, meaning the *differential* effect of MW across union and non-union workers is still identified. The state×year absorbs the common MW level, but cannot absorb how MW's effect differs between union and non-union workers within the same cell.

Intuitively, $\hat{\beta}_1$ asks: "do higher MW states have different outcomes?" — which state×year answers. $\hat{\beta}_3$ asks: "within a given state-year, do union workers respond differently to MW than non-union workers?" — state×year cannot answer, leaving $\hat{\beta}_3$ identified.

Interpret β_3 across outcomes. Under what conditions does unionization amplify or mitigate the effects of minimum wage changes? Note that the sign of β_3 may differ across outcomes — think carefully about the mechanisms.

$\hat{\beta}_3$ captures how union status moderates the effect of minimum wage changes across outcomes.
For log wages, $\hat{\beta}_3$ is near zero and insignificant across all specifications (≈ −0.0003 to −0.0005). Unionization neither amplifies nor dampens the wage effect of minimum wages — this makes sense, as union workers already have collectively bargained wages typically set above the minimum, so minimum wage increases bind less for them.
For employment, $\hat{\beta}_3$ is positive and stable (≈0.0098), meaning unions mitigate the disemployment effect of minimum wage increases. The mechanism is that collective

agreements stabilise worker-firm matches — layoffs require negotiation, making firms less able to immediately shed workers in response to rising labour costs.

For non-compliance, $\hat{\beta}_3$ is negative and stable (≈ -0.0050), meaning unions reduce the tendency for non-compliance to rise with minimum wages. This reflects unions' direct role in monitoring and enforcing wage floors through collective bargaining — effectively substituting for weak regulatory enforcement.

The asymmetry across outcomes is economically meaningful: unions do not change how much wages rise with the minimum wage, but they do cushion the labour market distortions — job losses and non-compliance — that minimum wage increases can generate.

Discuss the two identification concerns separately: (i) are firm-level union jumps exogenous to pre-existing wage and employment trends within the firm? (ii) are state-level minimum wage changes exogenous to state economic conditions? Also discuss the implications of treatment effect heterogeneity for two-way fixed effects estimators when the minimum wage is adopted in a staggered fashion across states.

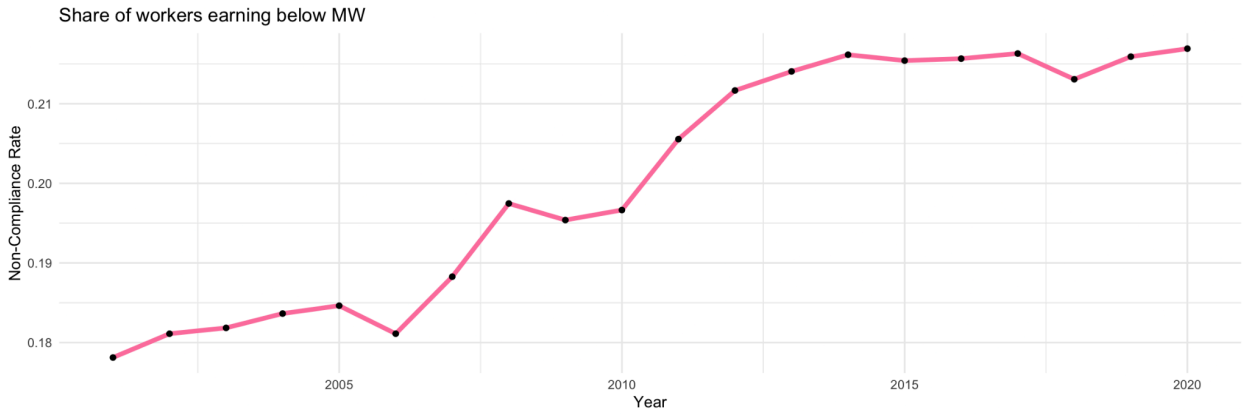
(i) Union jumps may be endogenous if workers unionize in response to deteriorating wages or employment prospects. The event-study plot showed visible pre-trends in wage dispersion before the union jump, particularly in the upper tail. A valid identification strategy requires parallel trends in the absence of unionization, the pre-trend evidence weakens this assumption.

(ii) Since the minimum wage is constant across all states and years in this particular dataset, the standard concerns about MW endogeneity (like states raising wages during booms, staggered adoption, negative TWFE weighting) don't apply here. There is no differential timing and no state-level policy endogeneity to worry about. All identification of $\hat{\beta}_1$ comes from individual-level variation in minimum wage exposure rather than cross-state policy variation.

When states adopt minimum wages at different times, the standard two-way fixed effects estimator creates a problem. It uses all available comparisons (like including already-treated states as controls for later-treated states). This is problematic because already-treated states are still experiencing their own treatment effects, making them invalid controls. If treatment effects are homogeneous this cancels out, but with heterogeneous effects (which holds as states differ in labour market conditions and adopt different MW levels) some comparisons receive negative implicit weights, potentially reversing the sign of the estimated effect entirely. We do not have this problem in this study as the state×year effects entirely absorb the minimum wage variation.

3.2 Non-Compliance

Define the non-compliance rate as the share of employed workers earning below the minimum wage. Compute and plot its evolution over time and across states. Do your estimates from Section 3.1 suggest that unionization reduces non-compliance? Relate this finding to the wage and employment results — are the three outcomes telling a consistent story?



The non-compliance rate rises steadily from approximately 18% in 2001 to over 21% by 2020, suggesting that over time a growing share of employed workers earn below the legal minimum. The source of the upward trend is possibly due to compositional changes in the workforce over time and the minimum wage rising over the sample period.

The regression estimates confirm that unionization significantly reduces non-compliance. The interaction coefficient $\hat{\beta}_3$ is negative and stable across all specifications (−0.0047 to −0.0057), and the total marginal effect of unions on non-compliance evaluated at MW=10 is −0.0337. This reflects unions directly monitoring and enforcing wage floors through collective bargaining, effectively substituting for weak regulatory enforcement.

The three outcomes tell a consistent and coherent economic story: higher minimum wages raise wages among employed workers but generate two distortions, namely, slight disemployment and rising non-compliance. Unionization mitigates both distortions without meaningfully changing the wage effect, suggesting unions act as an enforcement and stabilisation mechanism rather than simply amplifying the minimum wage's direct wage impact.

Appendix for U3

Step 1: Minimum Wage only

a) No fixed effects (OLS with worker controls only)

Variable	Model 1: Log Wage	Model 2: Employment	Model 3: Non-Compliance
Constant	2.413*** (0.0017)	0.9367*** (0.0021)	−0.3138*** (0.0028)
Min Wage	0.0452*** (0.0002)	−0.0028*** (0.0002)	0.0449*** (0.0003)
Age	0.0036*** (0.0000)	0.0000 (0.0000)	−0.0020*** (0.0000)
Female	−0.0316*** (0.0004)	−0.0101*** (0.0004)	0.0223*** (0.0006)
Education: HS or Less	−0.7371*** (0.0005)	−0.0301*** (0.0006)	0.3744*** (0.0008)
Education: Some College	−0.5164*** (0.0005)	−0.0107*** (0.0006)	0.1441*** (0.0008)
SE Type	IID	IID	IID
Observations	1,778,353	2,000,000	1,778,353
R^2	0.574	0.002	0.143

The omitted education category is BA or above — so all education coefficients are interpreted relative to degree holders
Model 2 has a very low R^2 (0.002), which is expected — a binary employment outcome is hard to explain with just demographics
Standard errors are IID (not clustered), which may understate uncertainty if there's within-group correlation

(b) Firm fixed effects: $Y_{if\,t} = \beta_1 MW_{st} + X_{it}\delta + \alpha_f + \varepsilon_{if\,t}$

Variable	Model 1: Log Wage	Model 2: Employment	Model 3: Non-Compliance
Min Wage	0.0485*** (0.0006)	−0.0020*** (0.0003)	0.0405*** (0.0005)
Female	−0.0317*** (0.0009)	−0.0102*** (0.0005)	0.0222*** (0.0009)
Age	0.0035*** (0.0000)	0.0000 (0.0000)	−0.0020*** (0.0000)

Education: HS or Less	−0.6903*** (0.0028)	−0.0313*** (0.0006)	0.3389*** (0.0027)
Education: Some College	−0.4868*** (0.0017)	−0.0115*** (0.0006)	0.1222*** (0.0022)
Firm FE	Yes	Yes	Yes
SE Clustering	by firm_id	by firm_id	by firm_id
Observations	1,778,353	2,000,000	1,778,353
R^2	0.728	0.004	0.211
Within R^2	0.638	0.002	0.123

R^2 jumps substantially (e.g. log wage goes from 0.574 to 0.728), showing firm fixed effects absorb a lot of wage variation — consistent with your earlier finding that between-firm differences matter
SE clustering by firm replaces IID errors — this is more credible since workers within the same firm are likely correlated
Education coefficients shrink slightly (e.g. HS or Less goes from −0.737 to −0.690 for log wages), suggesting part of the raw education gap was due to sorting of less-educated workers into lower-paying firms, not just within-firm pay differences
The Within R^2 is the more honest fit measure here — it tells you how much variation is explained after removing firm fixed effects

(c) Worker and firm fixed effects: $Y_{if\,t} = \beta_1 MW_{st} + X_{it}\delta + \alpha_i + \alpha_f + \varepsilon_{if\,t}$

Variable	Model 1: Log Wage	Model 2: Employment	Model 3: Non-Compliance
Min Wage	0.0489*** (0.0003)	−0.0020*** (0.0003)	0.0409*** (0.0004)
Age	0.0034*** (0.0000)	0.0000 (0.0000)	−0.0020*** (0.0000)
Female	−0.0308*** (0.0016)	−0.0103*** (0.0011)	0.0237*** (0.0018)
Education: HS or Less	−0.6605*** (0.0022)	−0.0323*** (0.0014)	0.3659*** (0.0022)
Education: Some College	−0.4636*** (0.0024)	−0.0118*** (0.0015)	0.1527*** (0.0023)
Worker FE	Yes	Yes	Yes

Firm FE	Yes	Yes	Yes
SE Clustering	by worker_id	by worker_id	by worker_id
Observations	1,778,353	2,000,000	1,778,353
R^2	0.865	0.054	0.330
Within R^2	0.398	0.000	0.035

R^2 jumps to 0.865 for log wages — worker FEs absorb enormous amounts of wage variation, reflecting persistent individual-level wage differences (ability, experience, etc.)
Within R^2 drops sharply (0.398 vs 0.638 in firm-FE only) — once you remove both worker and firm effects, the remaining within-match variation explained by observables is much smaller
Min wage coefficient stays remarkably stable across all three specifications (~0.045–0.049), which is reassuring — it suggests the result is robust to controlling for worker and firm heterogeneity
Education coefficients continue to shrink slightly (HS or Less: −0.737 → −0.690 → −0.661), consistent with some of the raw education gap being driven by sorting of less-educated workers into worse firms and lower-ability workers into lower education groups

(d) Worker, firm, and state×year fixed effects

Variable	Model 1: Log Wage	Model 2: Employment	Model 3: Non-Compliance
Min Wage	*dropped (collinear)*	*dropped (collinear)*	*dropped (collinear)*
Age	0.0034*** (0.0000)	0.0000 (0.0000)	−0.0020*** (0.0000)
Female	−0.0309*** (0.0016)	−0.0103*** (0.0011)	0.0237*** (0.0018)
Education: HS or Less	−0.6590*** (0.0022)	−0.0311*** (0.0014)	0.3639*** (0.0022)
Education: Some College	−0.4627*** (0.0024)	−0.0110*** (0.0015)	0.1514*** (0.0023)
Worker FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes
SE Clustering	by worker_id	by worker_id	by worker_id
Observations	1,778,353	2,000,000	1,778,353

R^2	0.865	0.054	0.330
Within R^2	0.357	0.000	0.029

Why min_wage drops: minimum wage varies at the state×year level. Once you include a dummy for every state×year combination, there is no remaining variation in min_wage to identify its effect — it's perfectly absorbed. This is a fundamental identification problem, not a coding error.

Practical implication: you cannot simultaneously control for all state-year shocks and estimate the effect of a state-year policy variable. This is the classic trade-off in minimum wage research.

Other coefficients are barely affected vs. Model c — age, female, and education coefficients are virtually identical, suggesting the state×year FE adds little beyond what worker+firm FEs already capture for those variables.

Within R^2 falls slightly (0.357 vs 0.398 in Model c), which is expected — the extra FEs absorb some additional residual variation.

UNIONS

(a) No fixed effects

Variable	Model 1: Log Wage	Model 2: Employment	Model 3: Non-Compliance
Constant	2.437*** (0.0031)	1.002*** (0.0037)	−0.3781*** (0.0049)
Min Wage	0.0413*** (0.0004)	−0.0113*** (0.0004)	0.0540*** (0.0006)
Union	0.0228*** (0.0037)	−0.0561*** (0.0045)	0.0146* (0.0059)
Age	0.0036*** (0.0000)	0.0000 (0.0000)	−0.0020*** (0.0000)
Female	−0.0316*** (0.0004)	−0.0101*** (0.0004)	0.0223*** (0.0006)
Education: HS or Less	−0.7366*** (0.0005)	−0.0298*** (0.0006)	0.3737*** (0.0008)
Education: Some College	−0.5160*** (0.0005)	−0.0105*** (0.0006)	0.1437*** (0.0008)
Min Wage × Union	−0.0004 (0.0004)	0.0081*** (0.0005)	−0.0047*** (0.0007)

Fixed Effects	None	None	None
SE Type	IID	IID	IID
Observations	1,778,353	2,000,000	1,778,353
R^2	0.574	0.002	0.144

The interaction term is the key here, it tells us whether the effect of minimum wage differs for union vs. non-union workers

For log wages, the interaction (−0.0004) is essentially zero and insignificant — meaning minimum wage raises wages similarly for union and non-union workers

For employment, the interaction (0.0081) is positive and significant — minimum wage increases actually reduce employment less for union workers, suggesting unions provide some buffer against job losses

For non-compliance, the interaction (−0.0047) is negative — union workers are less likely to be paid below the minimum when minimum wages rise, consistent with unions enforcing compliance through collective bargaining

(b) Firm fixed effects

Variable	Model 1: Log Wage	Model 2: Employment	Model 3: Non-Compliance
Min Wage	0.0414*** (0.0007)	−0.0149*** (0.0006)	0.0550*** (0.0008)
Union	0.0212*** (0.0053)	−0.0683*** (0.0051)	0.0234** (0.0072)
Age	0.0035*** (0.0000)	0.0000 (0.0000)	−0.0019*** (0.0000)
Female	−0.0317*** (0.0009)	−0.0102*** (0.0005)	0.0222*** (0.0009)
Education: HS or Less	−0.6900*** (0.0028)	−0.0311*** (0.0006)	0.3385*** (0.0027)
Education: Some College	−0.4866*** (0.0017)	−0.0114*** (0.0006)	0.1219*** (0.0022)
Min Wage × Union	−0.0003 (0.0006)	0.0097*** (0.0006)	−0.0058*** (0.0008)
Firm FE	Yes	Yes	Yes
SE Clustering	by firm_id	by firm_id	by firm_id

Observations	1,778,353	2,000,000	1,778,353
R^2	0.728	0.005	0.212
Within R^2	0.639	0.002	0.123

Min wage × union interaction is virtually unchanged — the log wage interaction stays near zero (−0.0003 vs −0.0004), the employment interaction rises slightly (0.0097 vs 0.0081), and the non-compliance interaction stays negative (−0.0058 vs −0.0047). The story is consistent across specifications.

The union main effect on employment becomes more negative (−0.0683 vs −0.0561) once firm FEs are added — suggesting part of the raw union-employment correlation was driven by firm-level differences, not union status itself

Within R^2 for log wages is high (0.639), consistent with the earlier firm-FE models — firm fixed effects absorb substantial wage variation

(c) Worker and firm fixed effects

Variable	Model 1: Log Wage	Model 2: Employment	Model 3: Non-Compliance
Min Wage	0.0419*** (0.0004)	−0.0150*** (0.0006)	0.0551*** (0.0008)
Union	0.0227*** (0.0036)	−0.0693*** (0.0051)	0.0236*** (0.0068)
Age	0.0034*** (0.0000)	0.0000 (0.0000)	−0.0020*** (0.0000)
Female	−0.0309*** (0.0016)	−0.0103*** (0.0011)	0.0237*** (0.0018)
Education: HS or Less	−0.6589*** (0.0022)	−0.0311*** (0.0014)	0.3638*** (0.0022)
Education: Some College	−0.4626*** (0.0024)	−0.0110*** (0.0015)	0.1513*** (0.0023)
Min Wage × Union	−0.0005 (0.0004)	0.0098*** (0.0006)	−0.0057*** (0.0008)
Worker FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
SE Clustering	by worker_id	by worker_id	by worker_id
Observations	1,778,353	2,000,000	1,778,353

R^2	0.865	0.054	0.330
Within R^2	0.400	0.001	0.036

The interaction term is remarkably stable across all three specifications — log wage interaction hovers around −0.0003 to −0.0005 (insignificant throughout), employment interaction stays around 0.0081–0.0098, and non-compliance around −0.0047 to −0.0058. This robustness strengthens the interpretation considerably

The union main effect on employment grows more negative as FEs are added (−0.0561 → −0.0683 → −0.0693), suggesting raw OLS was actually understating the negative union-employment relationship due to omitted firm/worker characteristics

R^2 jumps to 0.865 for log wages with two-way FEs, consistent with the earlier non-union models — worker heterogeneity dominates wage variation

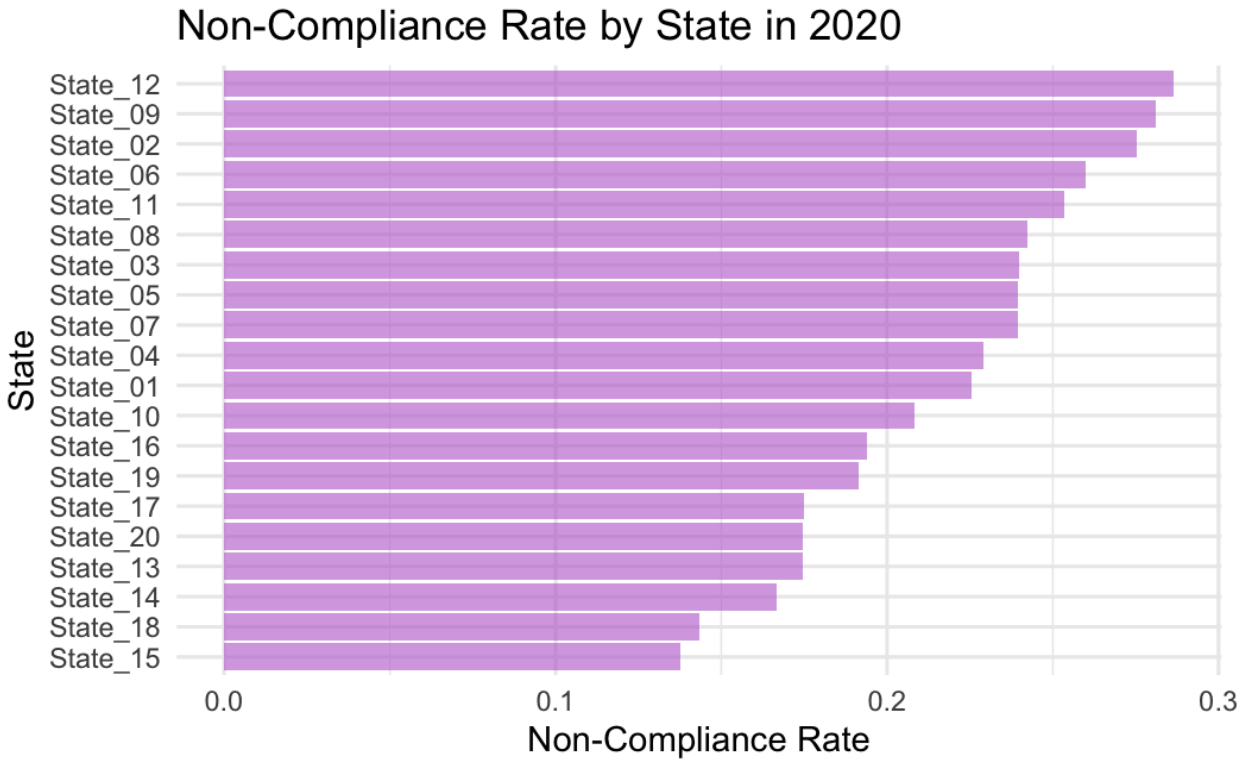
(d) Worker, firm, and state×year fixed effects

Variable	Model 1: Log Wage	Model 2: Employment	Model 3: Non-Compliance
Min Wage	*dropped (collinear)*	*dropped (collinear)*	*dropped (collinear)*
Union	0.0190*** (0.0041)	−0.0744*** (0.0080)	0.0072 (0.0100)
Age	0.0034*** (0.0000)	0.0000 (0.0000)	−0.0020*** (0.0000)
Female	−0.0309*** (0.0016)	−0.0103*** (0.0011)	0.0237*** (0.0018)
Education: HS or Less	−0.6590*** (0.0022)	−0.0311*** (0.0014)	0.3639*** (0.0022)
Education: Some College	−0.4627*** (0.0024)	−0.0110*** (0.0015)	0.1514*** (0.0023)
Min Wage × Union	−0.0001 (0.0005)	0.0105*** (0.0009)	−0.0039*** (0.0012)
Worker FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes
SE Clustering	by worker_id	by worker_id	by worker_id
Observations	1,778,353	2,000,000	1,778,353

R^2	0.865	0.055	0.330
Within R^2	0.357	0.001	0.029

The interaction term survives even though min_wage is dropped — this is the key advantage of the interaction design. Because union status varies within state-year cells, min_wage × union is not collinear with the state×year FEs and can still be identified

The interaction coefficients remain stable across all four specifications — employment (0.0081→0.0105) and non-compliance (−0.0047→−0.0039) tell a consistent story throughout

Non-compliance union main effect turns insignificant (0.0072, p>0.05) once state×year FEs are added, suggesting the raw union-compliance relationship was partly driven by state-level differences rather than union status itself

This specification is arguably your most credible — it controls for everything time-invariant at the worker and firm level, plus all state-year aggregate shocks, while still recovering the differential effect of minimum wages for union vs. non-union workers